

LEARNING MODULE RUBRIC

Module 0.

This module is designed to be an optical exercise for those lacking sufficient background in exoplanets. However, the following exercises are recommended:

1. **[BASIC]** Explore different parameters for the two widgets in the module. Write a reflection on how distance, proper motion, and solar flux influence a star's lightcurve. How does the lightcurve change when a planet is introduced?
2. **[INTERMEDIATE]** Think about the information a lightcurve provides and how this allows us to derive the transit depth and transit period of an exoplanet. What other parameters can we derive given an exoplanet's lightcurve? Work through the derivations.
3. **[ADVANCED]** Visit the source code for the exoplanet model in `setup.py`. What is the model missing? How can it be improved?

Module 1.

To receive credit for completing Module 1, students are expected to:

1. Complete all necessary functions
2. Successfully determine the orbital period of the example exoplanet selected and reflect on the accuracy of their findings
3. Work through the process of obtaining a periodogram and deriving the period, transit depth, transit time, and transit duration for at least ONE additional exoplanet

Module 2.

To receive credit for completing Module 2, students are expected to:

1. Complete all necessary functions
2. Answer all discussion questions
3. Produce all plots

Expansive learning

To gain a deeper understanding of modules 1 and 2, the following mini-projects are recommended

1. **[BASIC]** Determine the radius and distance from the host star for 5 exoplanets from **different** systems. Categorize these systems as being 1) Thin disk residents; 2) Thick disk residents; 3) Ex-situ systems; 4) Thick disk migrants
2. **[INTERMEDIATE]** Investigate the exoplanet population that resides in the thick disk in metallicity space. Where do they reside in the galaxy (look at their radial distances and scale heights)? Which planets are genuine thick disk occupants? Which ones have migrated to the thin disk? What differences between the two populations (if any) do you notice?
3. **[ADVANCED]** Look into where structures such as the spiral arms and stellar streams reside. Where do our exoplanet targets reside relative to these structures? Determine a list of exoplanet targets that reside nearby the spiral arms and/or stellar streams. Explore how this population differs from the generic thick and thin disk populations. Do you note any differences? What does this tell you about how the Galactic environment influences exoplanet formation?